

Enterprise Software Development

Lecture 08 - Agile Methodology





What we are discussing Today?

- ❑ What is Agile?
- ❑ Agile Manifesto
- ❑ Key Roles, Artifacts, Metrics of agile
- ❑ Agile Frameworks

What is Agile?

- ❑ Agile is an **iterative and incremental approach** to software development that emphasizes **flexibility**, **collaboration**, and **customer-centricity**.
- ❑ Unlike traditional waterfall methods, Agile methods aim to deliver working software in small, frequent releases rather than waiting for a final product.
- ❑ Agile methodologies prioritize responding to change and customer feedback, allowing teams to adapt to evolving requirements during development.

The Agile Manifesto

- The Agile Manifesto was created in 2001 by a group of software developers who sought to address the limitations of traditional software development approaches. It consists of **04 key values** and **12 principles** that guide Agile practices.

Key Values

1. Individuals and interactions over processes and tools:

Agile values human interactions and collaboration between team members more than relying solely on rigid processes and tools. Effective communication and teamwork are emphasized to build a successful project.

2. Working software over comprehensive documentation:

The primary measure of progress is delivering functional software to customers. While documentation is essential, Agile teams prioritize working software that provides tangible value over exhaustive documentation.

Key Values

3. Customer collaboration over contract negotiation:

Agile teams actively involve customers and stakeholders throughout the development process. Their collaboration ensures the delivered product meets customer needs and aligns with business goals.

4. Responding to change over following a plan:

Agile embraces change and understands that requirements can evolve during development. Teams are encouraged to adapt to changing circumstances, requirements, and priorities to deliver a successful product.

Principles

1. **Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.**

Agile teams prioritize delivering valuable features early and regularly to meet customer needs and gather feedback for continuous improvement.

2. **Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.**

Agile acknowledges that requirements are likely to change during a project, and teams are prepared to embrace and adapt to these changes to deliver maximum value.

3. **Deliver working software frequently, with a preference for shorter timescales.**

Agile promotes short development cycles, often referred to as sprints or iterations, to deliver functional software regularly and quickly.

4. **Business people and developers must work together daily throughout the project.**

Collaboration between business stakeholders and development teams is essential for shared understanding, feedback, and successful project delivery.

Principles

- 4. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.**

Agile empowers and trusts motivated individuals by providing them with a supportive environment to foster creativity, innovation, and self-organization.

- 5. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.**

While remote communication tools are valuable, Agile teams prioritize in-person communication to promote rapid information exchange and better understanding.

- 6. Working software is the primary measure of progress.**

The ultimate goal of Agile development is to deliver a functioning product increment at the end of each iteration or sprint.

- 7. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.**

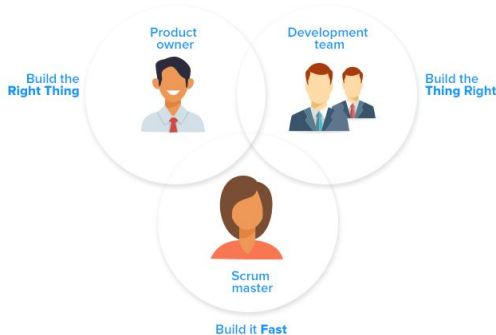
Agile emphasizes maintaining a sustainable pace for development teams to prevent burnout and ensure long-term productivity.

Principles

9. **Continuous attention to technical excellence and good design enhances agility.**
Agile teams focus on technical practices, code quality, and design principles to ensure the software remains maintainable, scalable, and adaptable.
10. **Simplicity—the art of maximizing the amount of work not done—is essential.**
Agile teams strive to keep things simple and avoid unnecessary complexity to minimize waste and deliver value more efficiently.
11. **The best architectures, requirements, and designs emerge from self-organizing teams.**
Agile fosters self-organizing teams that are empowered to make decisions and adapt to challenges effectively.
12. **At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.**
Agile promotes continuous improvement through regular retrospectives, where teams reflect on their processes, identify areas for improvement, and take action to enhance their performance.

Key Roles in Agile Teams

- ❑ **Product Owner:** Represents the customer and stakeholders, ensuring that the team works on the most valuable features first. They manage the Product Backlog and communicate the product vision to the team.
- ❑ **Scrum Master:** Serves as a facilitator, removing obstacles, and ensuring that the team adheres to Agile principles and practices. They help the team to be self-organizing and continuously improve.
- ❑ **Development Team:** A cross-functional, self-organizing group of professionals responsible for delivering the Increments of working software. They collaborate with the Product Owner and Scrum Master to meet the Sprint Goal.



Agile Artifacts

❏ **Product Backlog**

- The Product Backlog is a dynamic, prioritized list of all desired features, enhancements, and bug fixes for the product.
- The Product Owner is responsible for maintaining the Product Backlog, keeping it up-to-date, and ensuring that the highest-priority items are at the top.

❏ **Sprint Backlog:**

- The Sprint Backlog is a subset of the Product Backlog items selected for the current sprint.
- It represents the work that the Development Team has committed to completing during the sprint to achieve the Sprint Goal.

❏ **Increment**

- The Increment is the sum of all completed Product Backlog items at the end of a sprint.
- The Increment must be in a potentially releasable state, meeting the team's definition of done.

User Stories

- ❑ User stories are concise, simple, and easy-to-understand descriptions of desired functionality from the perspective of an end-user or customer.
- ❑ They serve as a means of capturing requirements and expressing features in a user-centered manner.
- ❑ User stories are a fundamental part of Agile project management and are used in various Agile frameworks like Scrum and Kanban.

A typical user story follows the format:

"As a [type of user], I want [an action or feature] so that [a reason or benefit]."

For example:

"As a website visitor, I want to create an account so that I can access exclusive content."

READ MORE: [User stories with examples and a template](#)

Agile Development Cycle

❑ **Sprint Planning**

The Product Owner and the Development Team collaborate to identify and select the Product Backlog items for the sprint. They define the Sprint Goal, which represents the purpose of the sprint and the value it aims to deliver.

❑ **Daily Standup**

A short, time-boxed meeting where the Development Team synchronizes their work. Each team member answers three questions: What did I do yesterday? What will I do today? Are there any impediments?

❑ **Sprint Review**

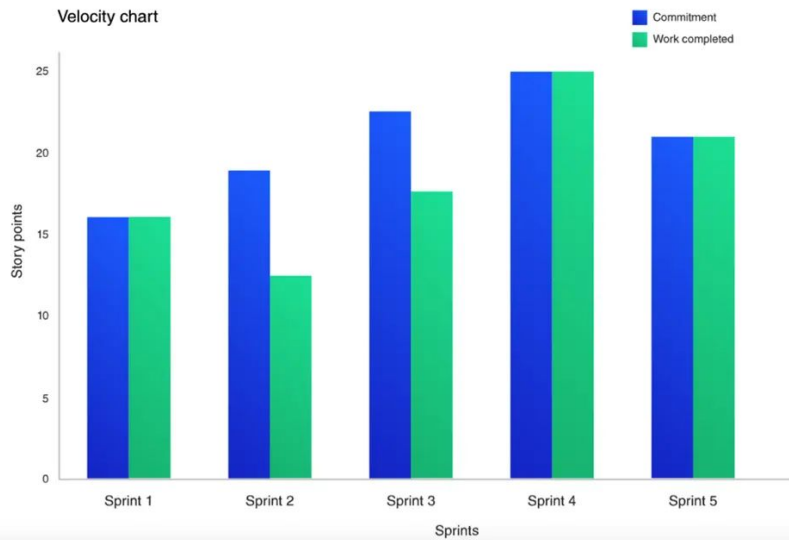
The Development Team presents the completed Increment to stakeholders and the Product Owner. Stakeholders provide feedback, and the Product Owner updates the Product Backlog based on the feedback.

❑ **Sprint Retrospective**

The Development Team reflects on the sprint and identifies what went well and what could be improved. They create action items to address improvement areas and enhance their processes.

Metrics in Agile

- ❑ **Velocity:** Velocity measures how much work a team completes in a sprint. It is calculated as the sum of story points or tasks completed during a sprint.



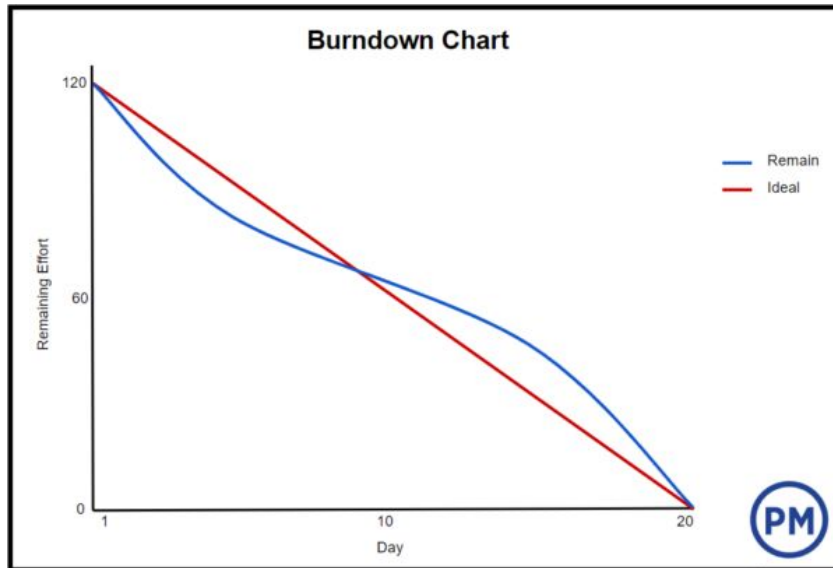
- ➔ Velocity is an indication of the average amount of Product Backlog turned into an Increment of product during Sprints.

Velocity graph helps,

- Predicting how much scope can be delivered by a specific date
- Predicting a date for a fixed amount of scope to be delivered
- Understanding limits while defining the amount of scope commit for next sprint

Metrics in Agile

- ❑ **Burndown Chart:** A visual representation of remaining work over the course of a sprint. It helps track progress and identify potential delays.



- Graphical representation of work left to do versus time
- Useful for predicting when all the work will be completed
- Ideally, burn-down chart should be checked by the team during the daily stand-up meeting

Burn-down chart helps,

- Monitoring the project scope creep
- Keeping the team running on schedule
- Comparing the planned work against the team progression

- ❑ Scrum
- ❑ Kanban
- ❑ Extreme Programming (XP)
- ❑ Lean Software Development

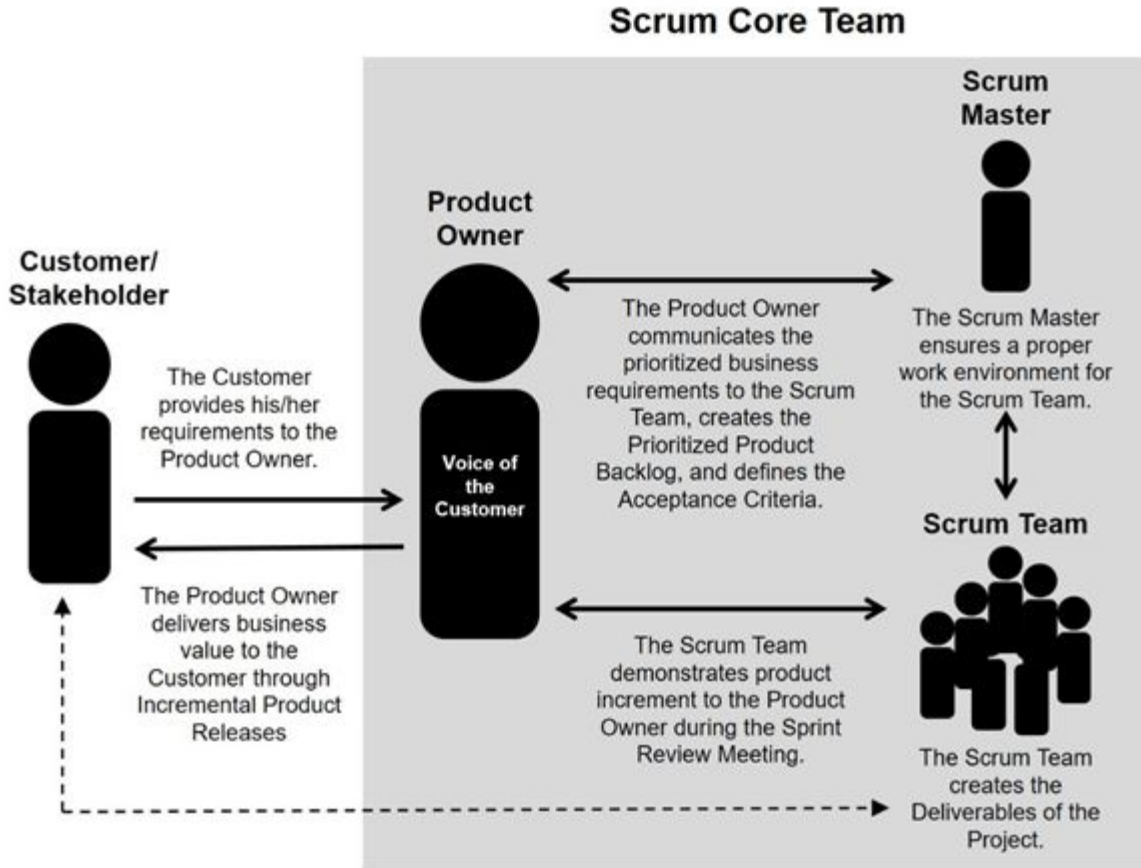
Agile Frameworks

- ❑ Scrum is one of the most widely used Agile frameworks. It provides a structured approach to software development, organizing work in fixed-length iterations called "**sprints**."
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Roles in Scrum

- Scrum Master: Facilitates the Scrum process, removes impediments, and coaches the team to improve.
- Product Owner: Represents the customers and stakeholders, prioritizes the backlog, and ensures the team builds the right product.
- Development Team: Self-organizing, cross-functional group responsible for delivering the Increment.

Scrum Framework



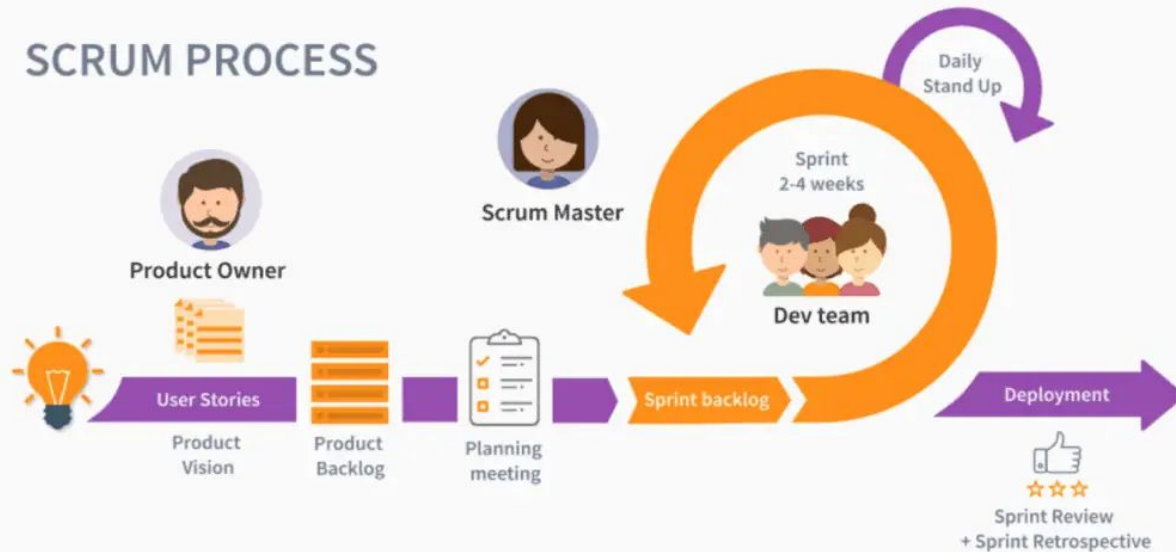
Scrum Framework

Scrum Events

- Sprint Planning: At the beginning of each sprint, the team collaborates with the Product Owner to select items from the Product Backlog and define the Sprint Goal.
- Daily Standup: A brief daily meeting where team members share progress, plans, and any obstacles they're facing.
- Sprint Review: At the end of the sprint, the team presents the Increment to stakeholders and gathers feedback.
- Sprint Retrospective: A reflection meeting where the team discusses what went well and what could be improved for the next sprint.

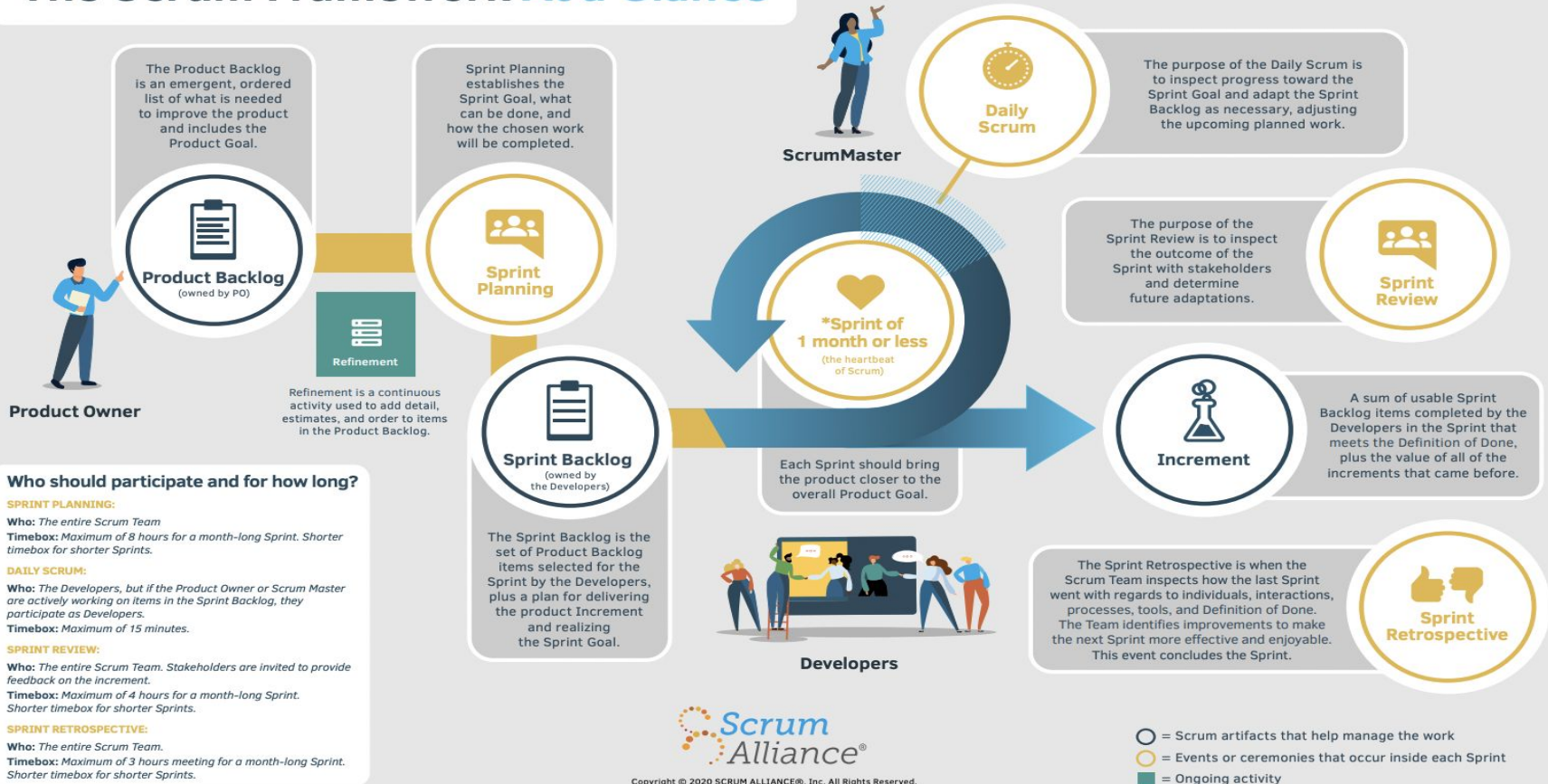
Scrum Framework

SCRUM PROCESS



Scrum Framework

The Scrum Framework At a Glance



- ❑ Kanban is a flexible Agile framework that **visualizes the workflow** and **limits work in progress (WIP)**. It is well-suited for continuous delivery and minimizing bottlenecks.
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Key concepts in Kanban

- Visual Kanban board: A board divided into columns representing different stages of work. Each task is represented by a card moving through the stages.
- Work in Progress (WIP) limits: Establishing a maximum number of items allowed in each column to prevent overloading the team.

Kanban Framework

Kanban Board

Backlog

To Do ②

Development ③

Testing ②

Deployment ①

Done

Explicit Policies

Cards must be prioritized and sized

Only product owner can add cards here

Any available developer can pull

Card is Done only when Code Review completed

Test cases must be documented before testing can start

Validation by PO

①

User Stories

Intangible

Defect

Fixed Date

Ongoing

Done

Ongoing

Done

F

G

D

C

E

B

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I



- ❑ Extreme Programming (XP) focuses on engineering practices and collaboration to produce high-quality software quickly.
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XP Values

- Communication: Foster open and frequent communication among team members and stakeholders.
 - Simplicity: Strive to keep things as simple as possible to avoid unnecessary complexity.
 - Feedback: Seek and utilize feedback from customers and team members to improve the product.
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XP Practices

- Test-Driven Development (TDD): Write tests before writing the actual code to ensure that code is functional and reliable.
- Pair Programming: Two developers work together at one workstation, promoting knowledge sharing and code quality.
- Continuous Integration: Frequently integrate code changes into a shared repository to catch integration issues early.

Extreme Programming

- ❑ Lean Software Development is based on Lean principles from manufacturing, focusing on delivering value to customers while eliminating waste and maximizing efficiency.
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Lean Principles

- Eliminate waste: Identify and remove non-value-adding activities in the development process.
- Amplify learning: Foster a culture of continuous learning and improvement throughout the organization.
- Decide as late as possible: Make decisions only when necessary to gather the most accurate and up-to-date information.
- Deliver as fast as possible: Aim to deliver working software as quickly as possible to gather feedback and validate assumptions.
- Empower the team: Encourage autonomy and trust in the development team to make decisions and solve problems effectively.
- Build integrity in: Focus on producing high-quality software that meets customer needs.

Lean Software Development

Assignment 08

1. Write a 04 page report excluding cover page and references on following topics.
 - Story points
 - Advantages of agile methodology
 - Challenges of agile methodology
 - Comparison of agile frameworks
2. Read more on Agile Development : <https://www.atlassian.com/agile>

Thanks!

Do you have any
 Questions?